In [3]:
```
# ========================================
# SALES PREDICTION FINAL PROJECT
# ========================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Styling for charts
sns.set_theme(style="whitegrid")

# ----------------------------------------
# STEP 1: CREATE  DATASET
# ----------------------------------------
np.random.seed(42)
n_samples = 1000

dates = pd.date_range(start='2024-01-01', periods=n_samples, freq='D')

advertising_spend = np.random.uniform(1000, 50000, n_samples)
discount_offered = np.random.uniform(0, 30, n_samples)
footfall = advertising_spend * 0.05 + np.random.normal(0, 200, n_samples)

# Sales formula
sales = 5000 + (0.8 * advertising_spend) - (150 * discount_offered) + (2.5 * footfall) + np.random.normal(0, 1500, n_samples)

df = pd.DataFrame({
    'Date': dates,
    'Advertising_Spend': advertising_spend,
    'Discount_Percent': discount_offered,
    'Footfall': footfall,
    'Sales': sales
})

print("--- Dataset Sample ---")
display(df.head(10))
print("\n")

# ----------------------------------------
# STEP 2: DATA PREPROCESSING & MODEL TRAINING
# ----------------------------------------
# Features (X) and Target (y)
X = df[['Advertising_Spend', 'Discount_Percent', 'Footfall']]
y = df['Sales']

# Train-Test Split (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Model Initialization & Training
model = LinearRegression()
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Model Evaluation
print("--- Model Performance ---")
print(f"R2 Score: {r2_score(y_test, y_pred):.4f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, y_pred)):.2f}\n")

# ----------------------------------------
# STEP 3: SAVE PREDICTIONS TO CSV
# ----------------------------------------
output_df = df.loc[X_test.index].copy() # Mapping test indices with dates
output_df['Predicted_Sales'] = y_pred

output_file_name = 'sales_predictions_output.csv'
output_df.to_csv(output_file_name, index=False)

# ----------------------------------------
# STEP 4: VISUALIZATIONS
# ----------------------------------------
fig, axes = plt.subplots(1, 3, figsize=(22, 6))

# Chart 1: Scatter Plot (Actual vs Predicted Sales)
sns.scatterplot(x=y_test, y=y_pred, ax=axes[0], color='blue', alpha=0.6)
axes[0].plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw=2) # Perfect fit line
axes[0].set_title('1. Actual vs Predicted Sales', fontsize=13, fontweight='bold')
axes[0].set_xlabel('Actual Sales', fontsize=11)
axes[0].set_ylabel('Predicted Sales', fontsize=11)

# Chart 2: Line Chart (Sales Trend Over Time)
df_sorted = df.sort_values('Date')
axes[2] # placeholder reference, let's plot on axes[1]
sns.lineplot(x='Date', y='Sales', data=df_sorted.head(100), ax=axes[1], color='green', linewidth=2)
axes[1].set_title('2. Sales Trend Over Time (First 100 Days)', fontsize=13, fontweight='bold')
axes[1].set_xlabel('Date', fontsize=11)
axes[1].set_ylabel('Sales', fontsize=11)
axes[1].tick_params(axis='x', rotation=30)

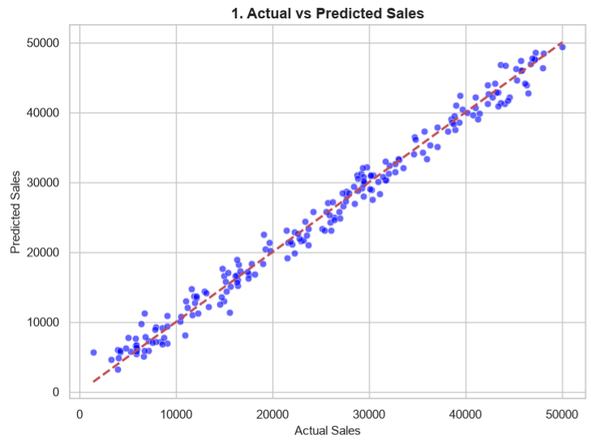
# Chart 3: Bar Chart (Feature Importance / Model Coefficients)
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_
})
sns.barplot(x='Feature', y='Coefficient', data=coefficients, ax=axes[2], palette='viridis', hue='Feature', legend=False)
axes[2].set_title('3. Feature Impact on Sales', fontsize=13, fontweight='bold')
axes[2].set_xlabel('Features', fontsize=11)
axes[2].set_ylabel('Coefficient Value', fontsize=11)

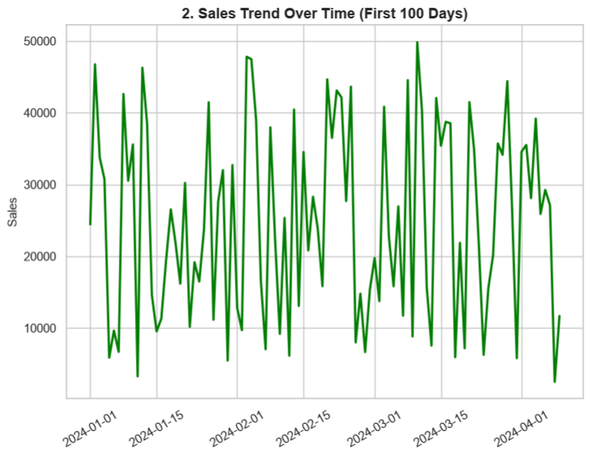
plt.tight_layout()
plt.show()
```

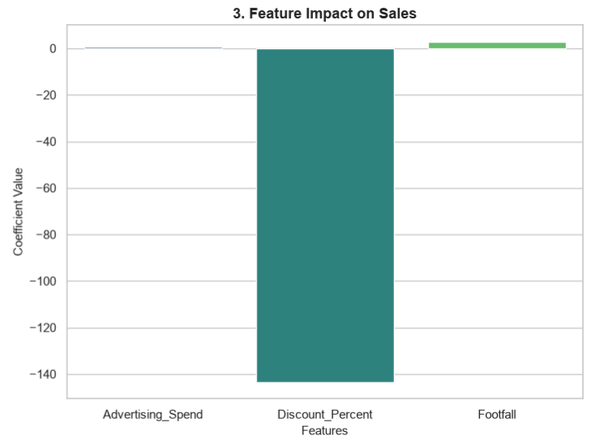
```
--- Dataset Sample ---
```

	Date	Advertising_Spend	Discount_Percent	Footfall	Sales
0	2024-01-01	19352.465824	5.553988	792.026774	24435.389177
1	2024-01-02	47585.001014	16.257028	2213.873980	46748.552764
2	2024-01-03	36867.703149	26.188375	1798.089379	33758.690607
3	2024-01-04	30334.265726	21.966747	1590.186388	30749.810190
4	2024-01-05	8644.913382	24.196834	614.962594	5870.190417
5	2024-01-06	8643.731496	19.763501	271.550785	9634.400269
6	2024-01-07	3846.096996	20.768297	490.842564	6680.471225
7	2024-01-08	43442.631143	25.475870	2117.906837	42613.896784
8	2024-01-09	30454.635575	7.490040	1518.458320	30515.504292
9	2024-01-10	35695.556312	14.682749	1635.335480	35570.902155

```
--- Model Performance ---
R2 Score: 0.9861
RMSE: 1560.45
```







In [ ]: